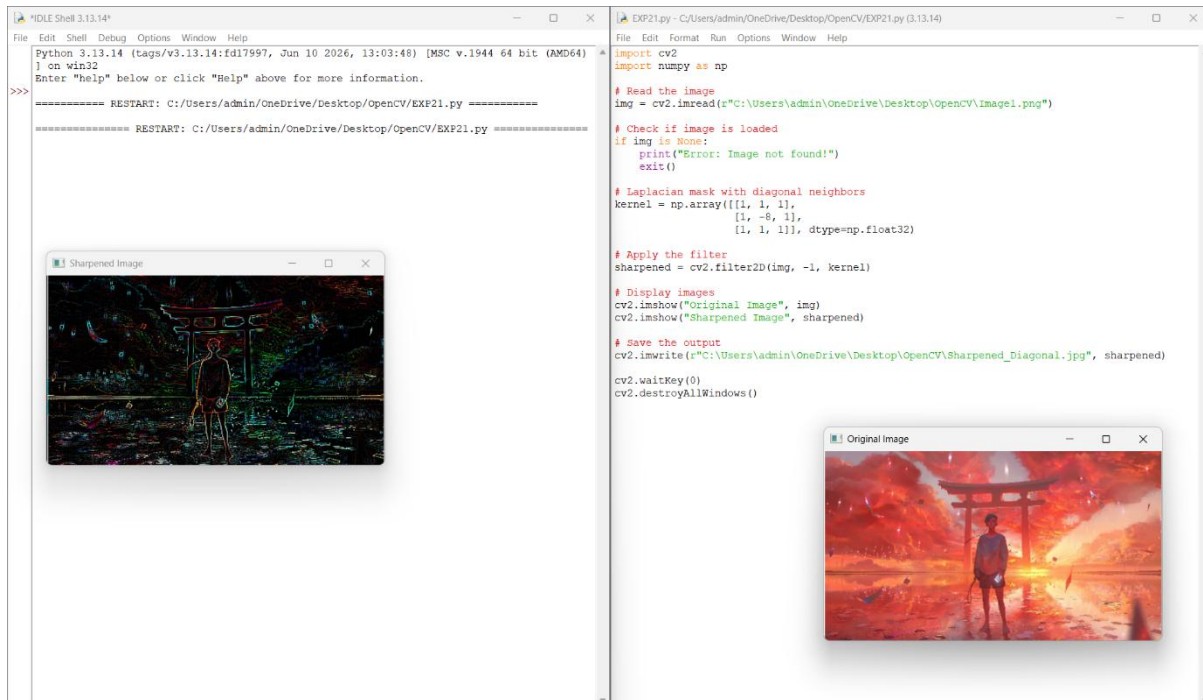
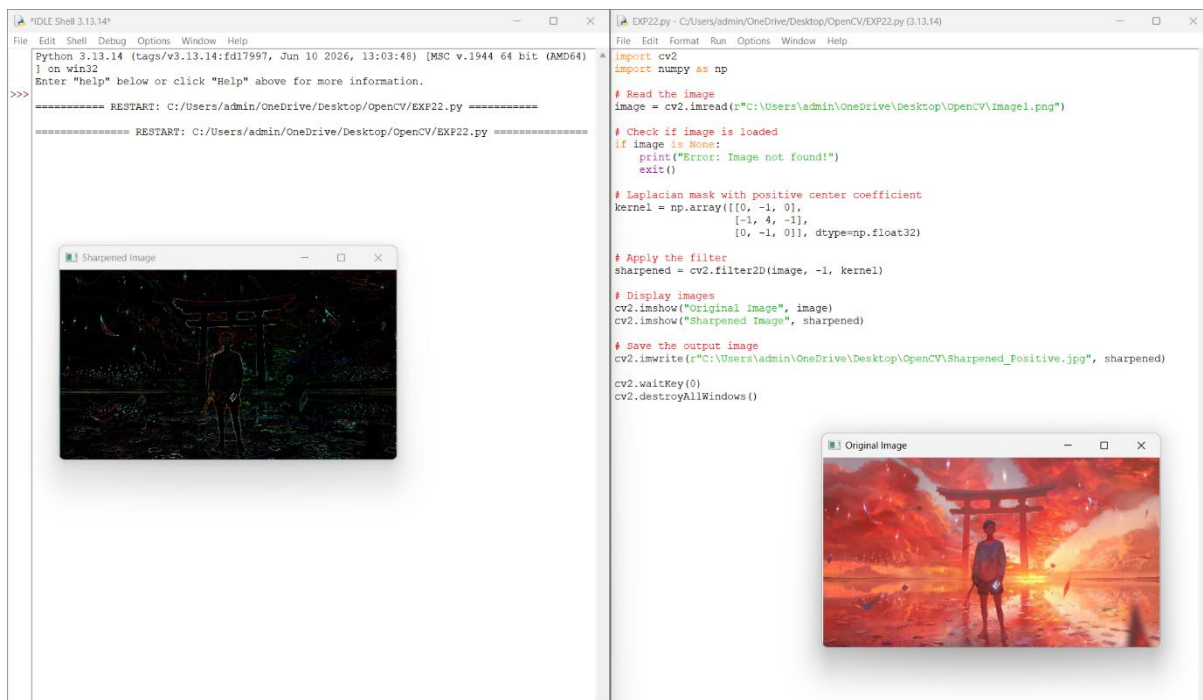


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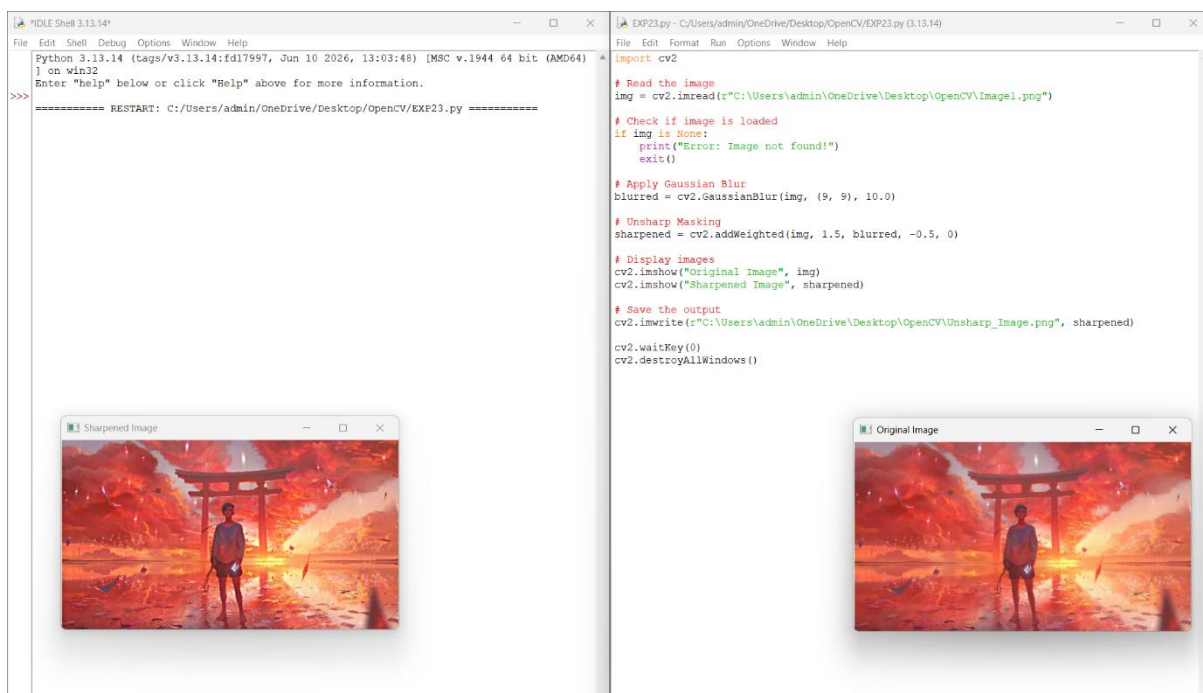
21.



22.



23.



The image displays a Windows desktop environment with three open windows:

- Terminal Window (Left):** Shows the execution of a Python script. The prompt is `>>>`. The script output includes:

```
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

===== RESTART: c:\Users\admin\oneDrive\Desktop\OpenCV\EXP24.py =====
```
- Image Window (Middle-Left):** Titled "High-Boost Sharpened Image", it displays a digital artwork of a person standing in a field of tall grass under a large, traditional Japanese torii gate. The image has been processed with high-boost sharpening, resulting in enhanced contrast and detail.
- Image Window (Middle-Right):** Titled "Original Image", it displays the same digital artwork as the first window, but in its original, unprocessed state.
- Code Editor Window (Right):** Shows the Python script `EXP24.py` with the following content:

```
import cv2
import numpy as np

# Function for High-Boost Filtering
def high_boost_filter(image, boost_factor):
    kernel_size = 3

    # Averaging kernel
    kernel = np.ones((kernel_size, kernel_size), dtype=np.float32) / (kernel_size * kernel_size)

    # Blur the image
    blur_image = cv2.filter2D(image, -1, kernel)

    # Convert to float for accurate calculation
    image = image.astype(np.float32)
    blur_image = blur_image.astype(np.float32)

    # High-Boost Mask
    sharpened = image + (image - blur_image) * boost_factor

    # Limit pixel values to 0-255
    sharpened = np.clip(sharpened, 0, 255).astype(np.uint8)

    return sharpened

# Read the image
image = cv2.imread(r"C:\Users\admin\oneDrive\Desktop\OpenCV\image1.png")

# Check if image is loaded
if image is None:
    print("Error: Image not found!")
    exit()

# Apply High-Boost Filtering
sharpened_image = high_boost_filter(image, 1.5)

# Display images
cv2.imshow("Original Image", image)
cv2.imshow("High-Boost Sharpened Image", sharpened_image)

# Save the output
cv2.imwrite(r"C:\Users\admin\oneDrive\Desktop\OpenCV\sharpened_image.png", sharpened_image)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

The image displays a screenshot of a Python script being executed in an IDE, demonstrating a gradient masking technique. The script is titled "EXP25.py" and is located at "C:\Users\admin\OneDrive\Desktop\OpenCV\EXP25.py (3.13.14)".

The script's execution output in the terminal shows the following steps:

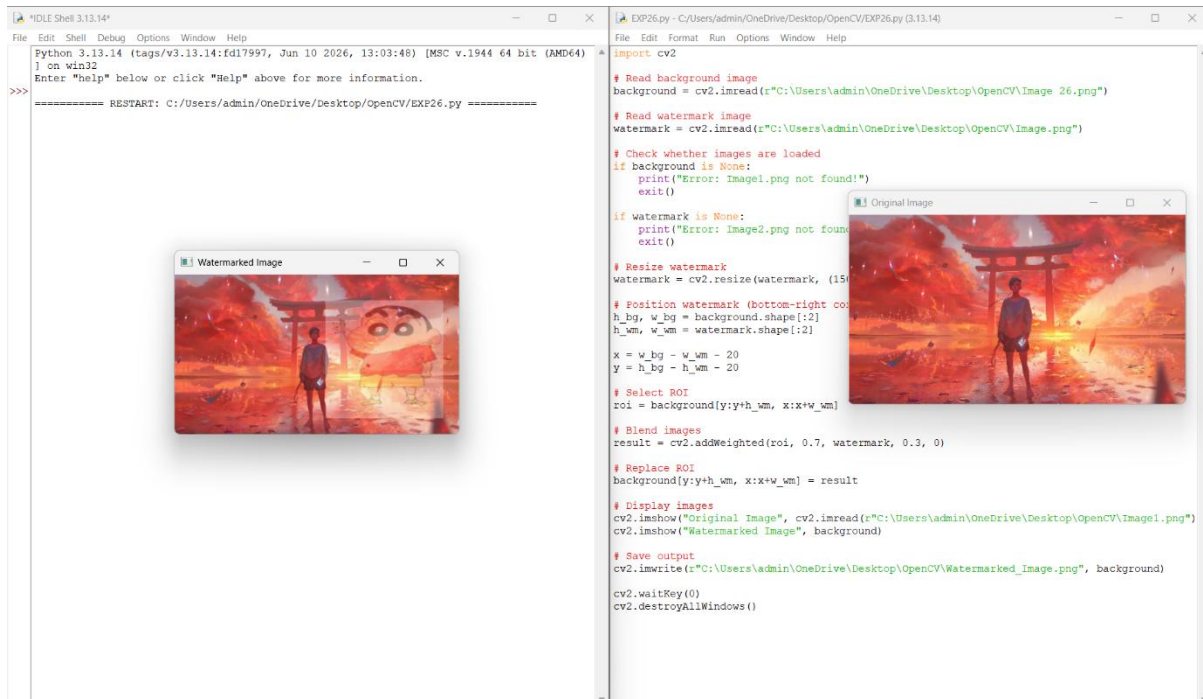
```
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\admin\OneDrive\Desktop\OpenCV\EXP25.py =====
===== RESTART: C:\Users\admin\OneDrive\Desktop\OpenCV\EXP25.py =====
```

The script code, visible in the editor, performs the following operations:

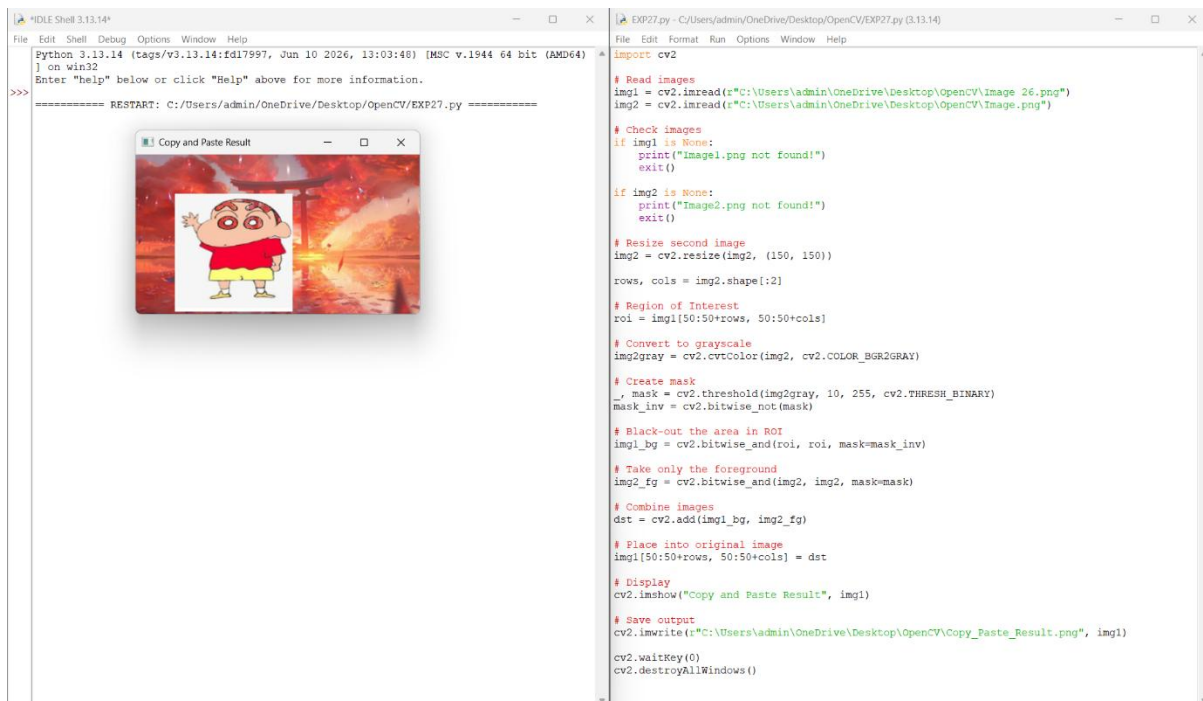
- Imports `cv2` and `numpy as np`.
- Reads the image `image = cv2.imread(r"C:\Users\admin\OneDrive\Desktop\OpenCV\image1.png")`.
- Checks if the image is loaded. If `image` is `None`, it prints an error and exits.
- Converts the image to grayscale: `gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)`.
- Calculates gradients: `gradient_x = cv2.Sobel(gray, cv2.CV_64F, 1, 0)` and `gradient_y = cv2.Sobel(gray, cv2.CV_64F, 0, 1)`.
- Computes the gradient mask: `gradient = cv2.subtract(gradient_x, gradient_y)`.
- Converts the gradient mask to 8-bit: `gradient = cv2.convertScaleAbs(gradient)`.
- Displays the images: `cv2.imshow("Original Image", image)` and `cv2.imshow("Gradient Masking", gradient)`.
- Saves the output: `cv2.imwrite(r"C:\Users\admin\OneDrive\Desktop\OpenCV\Gradient_Masking.jpg", gradient)`.
- Waits for a key press: `cv2.waitKey(0)`.
- Destroys all windows: `cv2.destroyAllWindows()`.

The script successfully generates a gradient mask, which is displayed in a window titled "Gradient Masking". This mask highlights the edges of the original image, showing a person standing in front of a torii gate. The original image is also displayed in a window titled "Original Image".

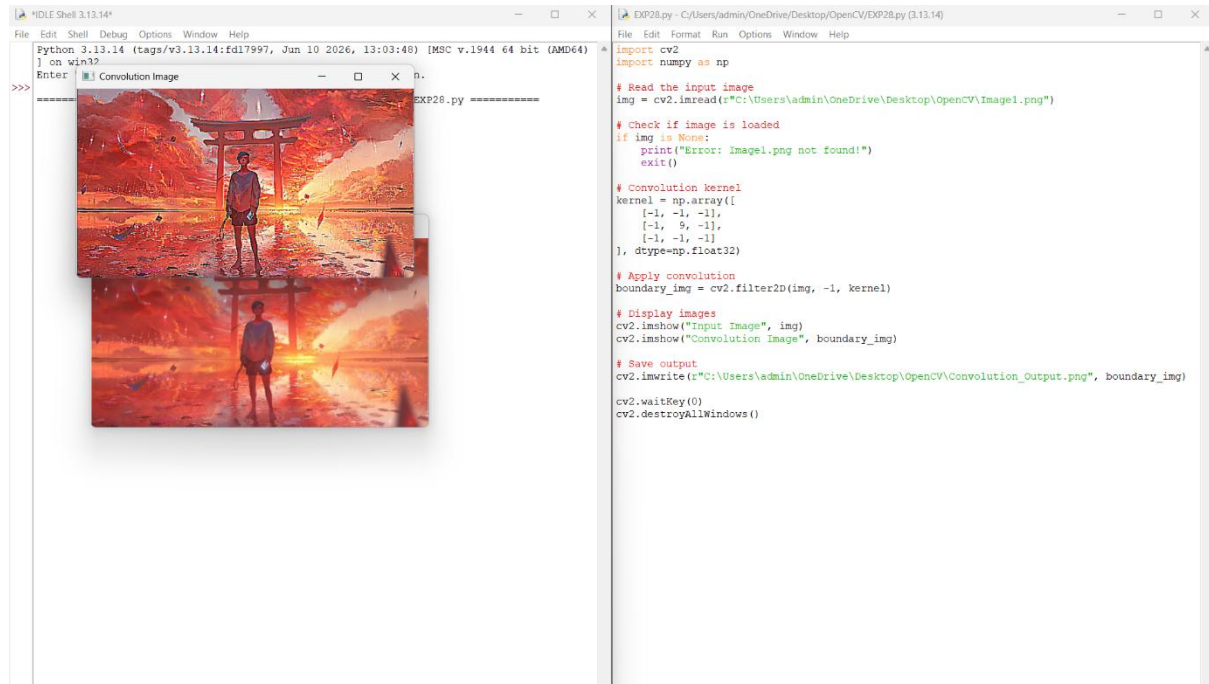
26.



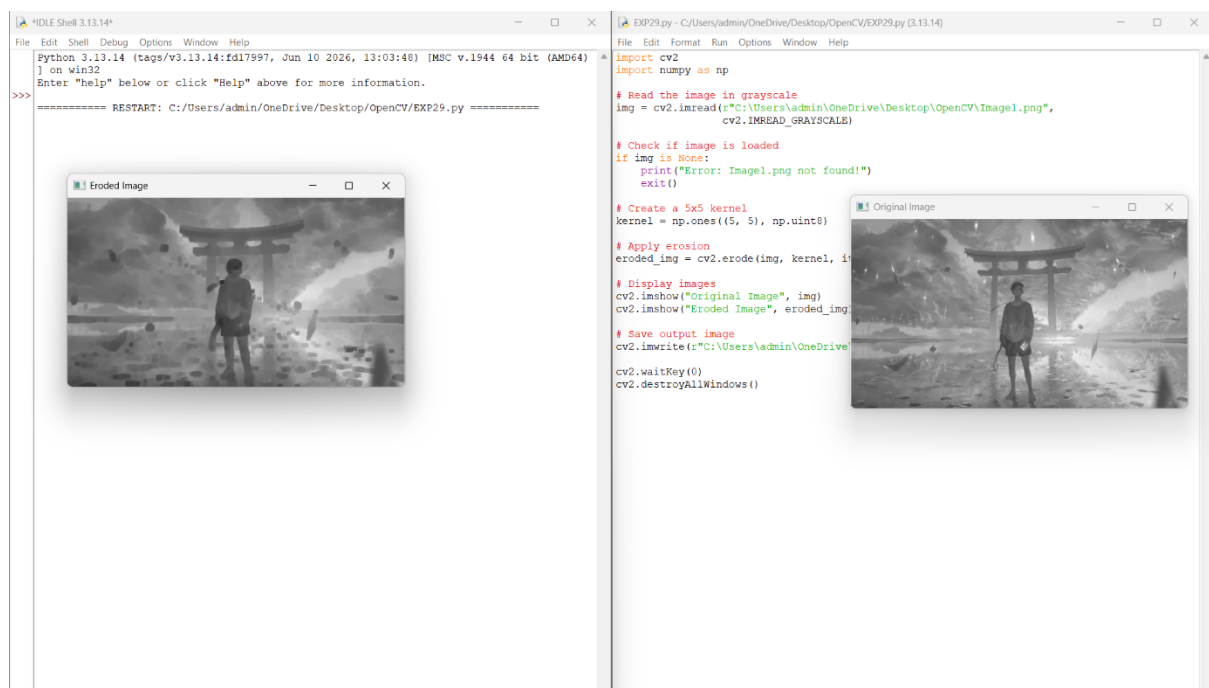
27.



28.



29.



30.

